

Jude Rifai

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

BS in Mechanical Engineering; Computer Science/Robotics Minors; GPA: 4.0/4.0

Expected December 2027

Relevant coursework: Thermodynamics, Dynamics, Data Structures/Algorithms

EXPERIENCE

Yellow Jacket Space Program Avionics Member

Jan. 2025 – Present

Georgia Institute of Technology

Atlanta, GA

- Developed embedded C/C++ firmware for sensor acquisition and real-time telemetry in flight systems, implementing structured data pipelines for hardware-software integration
- Built Python-based tooling for automated protocol analysis (SPI, I2C, CAN), including latency benchmarking, signal integrity evaluation, and debugging automation across subsystems
- Designed and implemented hardware validation scripts for PCB bring-up and system testing, reducing communication faults by 80% through software-driven diagnostics and iteration
- Modeled system dynamics in MATLAB and Python to simulate power distribution and thermal response under transient conditions, translating physical behavior into computational models for design optimization

Hypersonics Engineering Research Assistant

Jan. 2025 – Dec. 2025

Georgia Institute of Technology

Atlanta, GA

- Developed Python-based data processing pipelines for high-speed experimental systems, enabling structured analysis of pressure and temperature sensor streams
- Implemented signal processing and calibration routines (filtering, regression-based correction) for experimental sensors operating in extreme acoustic and thermal environments
- Built computational analysis tools to extract deformation and crack propagation metrics from experimental datasets, supporting validation of physical models

United Road Sustainability Engineering Intern

Nov. 2023 – Jan. 2024

United Road Services

Chattanooga, TN

- Built Python scripts to collect and organize real-world data on zero-emission vehicle platforms, including cost, performance, and reliability metrics from public and technical sources
- Automated extraction and structuring of regulatory requirements across 13 states using scripting and spreadsheet tools, creating a standardized dataset for analysis
- Developed a computational comparison model in Python to evaluate EV vs FCEV tradeoffs using aggregated real-world and regulatory data, supporting fleet selection decisions

UTC Biomechatronics Engineering Intern

Jan. 2023 – Apr. 2023

University of Tennessee at Chattanooga

Chattanooga, TN

- Developed SolidWorks CAD models and Python/MATLAB analysis tools for a linear-actuated assistive system, integrating mechanical design with computational evaluation workflows
- Built data processing scripts to analyze biomechanical testing results, enabling iterative system refinement under load conditions
- Applied computational analysis to experimental data to quantify performance improvements, achieving a 5% increase in load-bearing capacity through iterative design optimization

SKILLS

Computing: ROS, Mujoco, Gazebo, SolidWorks, C/C++, Python, MATLAB, Linux, CMake, Java, JavaScript, React

Manufacturing: Waterjet Cutting, Laser Cutting, Machining (Mill, Lathe), 3D Printing

Languages: English (native), Spanish (fluent), Arabic (fluent)

Clubs/Involvement: Robojackets, Startup Exchange, GT Climbing Club, Investments Committee

Interests: Karate, Climbing (V5), Jiu-Jitsu, Classical Guitar, Electric Guitar, Piano